

$$(z_0)^{1/n} \quad / \quad \underline{x^n - z_0 = 0}$$

Here $z_0 = a + ib$

$$\cos(2k\pi + \theta)$$

$$= \cos \theta$$

$$\begin{aligned}
 x &= (z_0)^{1/n} \\
 &= (a + ib)^{1/n} \\
 &= \left(r(\cos \theta + i \sin \theta) \right)^{1/n} \\
 &= r^{1/n} \left[\cos(2k\pi + \theta) + i \sin(2k\pi + \theta) \right]^{1/n} = r^{1/n} \left(e^{i(2k\pi + \theta)} \right)^{1/n} \\
 &= r^{1/n} \left[\cos\left(\frac{2k\pi + \theta}{n}\right) + i \sin\left(\frac{2k\pi + \theta}{n}\right) \right] = r^{1/n} e^{i\left(\frac{2k\pi + \theta}{n}\right)}
 \end{aligned}$$

$k = 0, 1, 2, \dots, n-1$

$$z_0 = a + ib$$

To find $(z_0)^{1/n} / x^n - z_0 = 0$

$$x^n - z_0 = 0$$

$$x = (z_0)^{1/n}$$

$$= (a + ib)^{1/n}$$

$$= [r(\cos \theta + i \sin \theta)]^{1/n}$$

$$= (r e^{i\theta})^{1/n}$$

$$= r^{1/n} [\cos(2k\pi + \theta) + i \sin(2k\pi + \theta)]^{1/n} = r^{1/n} (e^{i(2k\pi + \theta)})^{1/n}$$

$$= r^{1/n} \left[\cos\left(\frac{2k\pi + \theta}{n}\right) + i \sin\left(\frac{2k\pi + \theta}{n}\right) \right] = r^{1/n} \left[e^{i\frac{(2k\pi + \theta)}{n}} \right]$$

$$k = 0, 1, 2, \dots, n-1$$

$$k=0 \quad x_0 = r^{1/n} \left[\cos\left(\frac{\theta}{n}\right) + i \sin\left(\frac{\theta}{n}\right) \right]$$

$$k=1 \quad x_1 = r^{1/n} \left[\cos\left(\frac{2\pi + \theta}{n}\right) + i \sin\left(\frac{2\pi + \theta}{n}\right) \right]$$

$$k=n-1 \quad x_{n-1} =$$

$$k=n \quad x_n = r^{1/n} \left[\cos\left(\frac{2n\pi + \theta}{n}\right) + i \sin\left(\frac{2n\pi + \theta}{n}\right) \right]$$

$$= r^{1/n} \left(\cos\left(2\pi + \frac{\theta}{n}\right) + i \sin\left(2\pi + \frac{\theta}{n}\right) \right)$$

To find $(1)^{1/n}$

$$x = (1)^{1/n}$$

$$= (1 e^{i0})^{1/n}$$

$$= (e^{i(2k\pi+0)})^{1/n}$$

$$= (e^{i \frac{2k\pi}{n}})$$

$$(\cos 0 + i \sin 0)^{1/n}$$

$$(\cos(2k\pi) + i \sin(2k\pi))^{1/n}$$

$$k = 0, 1, 2, 3 \dots n-1$$

$$k=0 \quad x_0 = 1$$

$$k=1 \quad x_1 = e^{i \frac{2\pi}{n}} = \omega$$

$$k=2 \quad x_2 = e^{i \frac{4\pi}{n}} = (e^{i \frac{2\pi}{n}})^2 = \omega^2$$

$$k=n-1 \quad x_{n-1} = \omega^{n-1}$$

$1, \omega, \omega^2, \dots, \omega^{n-1}$ are n^{th} roots of unity

$$\omega^n = (e^{i \frac{2\pi}{n}})^n$$

$$= e^{i 2\pi}$$

$$= \cos 2\pi + i \sin 2\pi$$

$$= 1$$

$$\text{Sum } 1 + \omega + \omega^2 + \dots + \omega^{n-1} = \frac{1 - \omega^n}{1 - \omega}$$

$$\omega \neq 1$$

$$= \frac{1-1}{1-\omega} = 0$$

$$1 - x^2 = (1-x)(1+x)$$

$$1 - x^3 = (1-x)(1+x+x^2)$$

$$1 - x^4 = (1-x)(1+x+x^2+x^3)$$

$$1 - x^n = (1-x)(1+x+\dots+x^{n-1})$$

Continued product

$$1 \cdot \omega \cdot \omega^2 \dots \omega^{n-1} = \omega^{1+2+\dots+n-1}$$

$$1+2+\dots+n = \frac{n(n+1)}{2}$$

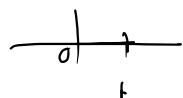
$$= \omega^{\frac{(n+1)n}{2}} = (\omega^n)^{\frac{n+1}{2}} = (1)^{\frac{n+1}{2}} = 1$$

To find n^{th} roots of unity or $(1)^{1/n}$ or $x^n - 1 = 0$

$$x = (1)^{1/n}$$

$$= (\cos 0 + i \sin 0)^{1/n}$$

$$= [\cos(2k\pi) + i \sin(2k\pi)]^{1/n}$$



By De Moivre's thm

$$= \cos\left(\frac{2k\pi}{n}\right) + i \sin\left(\frac{2k\pi}{n}\right) \quad k=0, 1, 2, \dots, n-1.$$

$$= e^{i\left(\frac{2k\pi}{n}\right)} \quad k=0, 1, 2, \dots, n-1$$

$$k=0, \quad x_0 = 1$$

$$k=1, \quad x_1 = e^{i\frac{2\pi}{n}} = \omega$$

$$k=2, \quad x_2 = e^{i\frac{4\pi}{n}} = \left(e^{i\frac{2\pi}{n}}\right)^2 = \omega^2$$

$\vdots$

$\vdots$

$$k=n-1 \quad x_{n-1} = \omega^{n-1}$$

$$\begin{aligned} \omega^n &= \left(e^{i\frac{2\pi}{n}}\right)^n = e^{i2\pi} \\ &= \cos 2\pi + i \sin 2\pi = 1 \\ \Rightarrow \omega^n &= 1 \end{aligned}$$

① Sum of n^{th} roots of unity -

$$\frac{1 + \omega + \omega^2 + \dots + \omega^{n-1}}{1 - \omega} = \frac{1 - \omega^n}{1 - \omega} = \frac{1 - 1}{1 - \omega} = 0$$

$$\begin{aligned} (1 - x^2) &= (1 - x)(1 + x) \\ 1 - x^3 &= (1 - x)(1 + x + x^2) \\ (1 - x^4) &= (1 - x)(1 + x + x^2 + x^3) \\ &\vdots \\ (1 - x^n) &= (1 - x)(1 + x + x^2 + \dots + x^{n-1}) \end{aligned}$$

Note: Continued Product is product of all roots

$$1 \cdot \omega \cdot \omega^2 \cdot \dots \cdot \omega^{n-1} = \omega^{1+2+\dots+(n-1)} = \omega^{\frac{n(n-1)}{2}} = (\omega^n)^{\frac{n-1}{2}} = 1^{\frac{n-1}{2}} = 1$$

$$\sum n = \frac{n(n+1)}{2} \quad \sum n-1 = \frac{(n-1)n}{2}$$

$$\begin{aligned}
 \lambda &= (1)^{1/3} \\
 &= (e^{i0})^{1/3} \\
 &= (e^{i2k\pi})^{1/3} = e^{i\frac{2k\pi}{3}} \\
 &\quad k=0, 1, 2
 \end{aligned}$$

$k=0, \quad \lambda_0 = 1$
 $k=1, \quad \lambda_1 = e^{i\frac{2\pi}{3}} = \omega$
 $k=2, \quad \lambda_2 = e^{i\frac{4\pi}{3}} = \omega^2$

$1, \omega, \omega^2$ are ^{cube} roots of unity
 $\therefore \omega^3 = 1$

$$(1-\omega)^6 = \left[(1-\omega)^2 \right]^3 = (1-2\omega+\omega^2)^3$$

$$\text{as } \underline{1+\omega+\omega^2} = \frac{1-\omega^3}{1-\omega} = \frac{1-1}{1-\omega} = 0$$

$$\Rightarrow 1+\omega^2 = -\omega$$

$$(1-\omega)^6 = (1+\omega^2-2\omega)^3 = (-\omega-2\omega)^3 = (-3\omega)^3 = -27\omega^3 = -27$$

Find the cube roots of $(1 - \cos\theta - i \sin\theta)$.

$$\begin{cases} 1 + \cos 2\theta = 2 \cos^2 \theta \\ 1 - \cos 2\theta = 2 \sin^2 \theta \\ \sin 2\theta = 2 \sin \theta \cos \theta \end{cases}$$

$$(1 - \cos\theta - i \sin\theta)^{1/3}$$

$$= (2 \sin^2 \theta / 2 - i 2 \sin \theta / 2 \cos \theta / 2)^{1/3}$$

$$= (2 \sin \theta / 2)^{1/3} \left(\sin \frac{\theta}{2} - i \cos \frac{\theta}{2} \right)^{1/3}$$

$$= (2 \sin \theta / 2)^{1/3} \left[\cos \left(\frac{\pi}{2} - \frac{\theta}{2} \right) - i \sin \left(\frac{\pi}{2} - \frac{\theta}{2} \right) \right]^{1/3}$$

Consider $\frac{\pi}{2} - \frac{\theta}{2} = \phi$

$$= (2 \sin \theta / 2)^{1/3} [\cos \phi - i \sin \phi]^{1/3}$$

Hw $x^5 + 1 = 0$

$$= (2 \sin \theta / 2)^{1/3} \left[\cos (2k\pi + \phi) - i \sin (2k\pi + \phi) \right]^{1/3}$$

$$= (2 \sin \theta / 2)^{1/3} \left[\cos \left(\frac{2k\pi + \phi}{3} \right) - i \sin \left(\frac{2k\pi + \phi}{3} \right) \right]$$

$k = 0, 1, 2$

$$= (2 \sin \theta / 2)^{1/3} \left[\cos \left(\frac{2k\pi + \left(\frac{\pi - \theta}{2} \right)}{3} \right) - i \sin \left(\frac{2k\pi + \left(\frac{\pi - \theta}{2} \right)}{3} \right) \right]$$

$$\left. \begin{aligned} 2k\pi + \frac{\pi - \theta}{2} \\ 4k\pi + \pi - \theta \end{aligned} \right\} = \frac{(4k+1)\pi - \theta}{2}$$

$$\begin{aligned} x^5 + 1 &= 0 \\ x &= (-1)^{1/5} \\ &= (10e^{i\pi})^{1/5} \end{aligned}$$

$$\begin{aligned} x^5 + 1 &= 0 \Rightarrow x = (-1)^{1/5} \\ &= (e^{i\pi})^{1/5} \end{aligned}$$

z^n

$$z^n - z_0 = 0$$

$$z = (z_0)^{1/n}$$

$$= (r e^{i\theta})^{1/n}$$

$$= (r e^{i(2k\pi + \theta)})^{1/n}$$

$$= \left(r e^{i \frac{(2k\pi + \theta)}{n}} \right)$$

$$k = 0, 1, \dots, n-1$$

Find all the values of $\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^{3/4}$ and show that their continued product is 1.

$$\frac{1}{2} + i\frac{\sqrt{3}}{2}$$

$$r$$

$$1$$

$$\theta$$

$$\theta = \alpha = \frac{\pi}{3}$$

$$\begin{aligned} \left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^{3/4} &= \left[\left(1 \cdot e^{i\pi/3}\right)^3 \right]^{1/4} \\ &= \left(e^{i\pi}\right)^{1/4} = \left(e^{i(2k\pi + \pi)}\right)^{1/4} \\ &= e^{i\left(\frac{2k\pi + \pi}{4}\right)} \quad k = 0, 1, 2, 3 \end{aligned}$$

$$k=0$$

$$x_0 = e^{i\pi/4}$$

$$k=1$$

$$x_1 = e^{i3\pi/4}$$

$$k=2$$

$$x_2 = e^{i5\pi/4}$$

$$k=3$$

$$x_3 = e^{i7\pi/4}$$

Continued product

$$= x_0 x_1 x_2 x_3$$

$$= e^{i\pi/4} e^{i3\pi/4} e^{i5\pi/4} e^{i7\pi/4}$$

$$= e^{i4\pi} = \cos 4\pi + i \sin 4\pi = 1$$

Solve: $x^7 + x^4 + x^3 + 1 = 0$

$$x^4(x^3 + 1) + (x^3 + 1) = 0$$

$$(x^3 + 1)(x^4 + 1) = 0$$

$$x^3 + 1 = 0$$

$$x = (-1)^{1/3}$$

$$= (e^{i\pi})^{1/3}$$

$$= \left(e^{i(2k\pi + \pi)} \right)^{1/3}$$

$$= e^{i \frac{(2k\pi + \pi)}{3}}$$

$$k = 0, 1, 2$$

$$x^4 + 1 = 0$$

$$x = (-1)^{1/4}$$

$$= (e^{i\pi})^{1/4}$$

$$= \left(e^{i(2k\pi + \pi)} \right)^{1/4}$$

$$= e^{i \frac{(2k\pi + \pi)}{4}}$$

$$k = 0, 1, 2, 3$$

• Solve: $x^4 + x^3 + x^2 + x + 1 = 0$

$$x^2 - 1 = (x-1)(x+1)$$

$$x^3 - 1 = (x-1)(x^2 + x + 1)$$

$$x^4 - 1 = (x-1)(x^3 + x^2 + x + 1)$$

$$x^5 - 1 = (x-1)(x^4 + x^3 + x^2 + x + 1) = 0$$

we'll solve

$$x^5 - 1 = 0$$

$$\Rightarrow x = (1)^{1/5} = (e^{i0})^{1/5}$$

$$= (e^{i(2k\pi)})^{1/5} = e^{i\frac{2k\pi}{5}}$$

$$k = 0, 1, 2, 3, 4$$

$$k=0 \quad x_0 = 1$$

$$k=1 \quad x_1 = e^{i2\pi/5} = \omega$$

$$k=2 \quad x_2 = \omega^2$$

$$k=3 \quad x_3 = \omega^3$$

$$k=4 \quad x_4 = \omega^4$$

as $(x-1)(x^4 + x^3 + x^2 + x + 1)$ are factors of $x^5 - 1$.

& $x=1$ is obviously root of $x-1=0$

$\therefore \omega, \omega^2, \omega^3, \omega^4$ are roots of

$$x^4 + x^3 + x^2 + x + 1 = 0$$

Find the roots common to $x^4 + 1 = 0$ and $x^6 - i = 0$.

$$x^4 + 1 = 0$$

$$x = (-1)^{1/4}$$

$$x = e^{i \left(\frac{2k\pi + \pi}{4} \right)}$$

$$k = 0, 1, 2, 3$$

$$x_0 = e^{i\pi/4}$$

$$x_1 = e^{i3\pi/4}$$

$$x_2 = e^{i5\pi/4}$$

$$x_3 = e^{i7\pi/4}$$

Common roots are

$$e^{i3\pi/4}$$

$$e^{i7\pi/4}$$

$$x^6 - i = 0$$

$$x = (i)^{1/6}$$

$$= \left(e^{i\pi/2} \right)^{1/6}$$

$$= e^{i \left(\frac{2k\pi + \pi/2}{6} \right)}$$

$$= e^{i \left(\frac{4k\pi + \pi}{12} \right)}$$

$$k = 0, 1, 2, 3, 4, 5$$

$$x_0 = e^{i\pi/12}$$

$$x_1 = e^{i5\pi/12}$$

$$x_2 = e^{i9\pi/12}$$

$$= e^{i3\pi/4}$$

$$x_3 = e^{i13\pi/12}$$

$$x_4 = e^{i17\pi/12}$$

$$x_5 = e^{i21\pi/12} = e^{i7\pi/4}$$

- If $\alpha, \alpha^2, \alpha^3, \alpha^4$, are the roots of $x^5 - 1 = 0$, find them & show that $(1 - \alpha)(1 - \alpha^2)(1 - \alpha^3)(1 - \alpha^4) = 5$.

$$x^5 = 1 \Rightarrow x = (\pm 1)^{1/5}$$

$$x = (e^{i0})^{1/5}$$

$$= (e^{i2k\pi})^{1/5} = e^{i\frac{2k\pi}{5}} \quad k=0, 1, 2, 3, 4$$

$$x_0 = 1$$

$$x_0 = e^{i2\pi/5} = \alpha$$

$$x_1 = e^{i4\pi/5} = \alpha^2$$

$$\left| \begin{array}{l} x_2 = e^{i6\pi/5} = \alpha^3 \\ x_3 = e^{i8\pi/5} = \alpha^4 \end{array} \right.$$

$$x^5 - 1 = (x-1)(x^4 + x^3 + x^2 + x + 1)$$

$$\Rightarrow \cancel{(x-1)} (x-\alpha) (x-\alpha^2) (x-\alpha^3) (x-\alpha^4) = \cancel{(x-1)} (x^4 + x^3 + x^2 + x + 1)$$

$$\begin{array}{l} \text{put } x=1 \\ (1-\alpha) (1-\alpha^2) (1-\alpha^3) (1-\alpha^4) = 5 \end{array}$$

$$x^4 = i$$

- Solve the equation $z^4 = i(z-1)^4$ and show that the real part of all the roots is $1/2$.

$$\frac{z^4}{(z-1)^4} = i \Rightarrow \left(\frac{z}{z-1}\right)^4 = i$$

$$\frac{z}{z-1} = (i)^{1/4} = (e^{i\pi/2})^{1/4} = (e^{i(2k\pi + \pi/2)})^{1/4}$$

$$= e^{i(4k\pi + \pi)/8}$$

$$k = 0, 1, 2, 3$$

$$\frac{z}{z-1} = e^{i\theta} = \frac{\cos\theta + i\sin\theta}{1}$$

$$\frac{z}{(z-1)-z} = \frac{\cos\theta + i\sin\theta}{1 - \cos\theta - i\sin\theta}$$

$$\Rightarrow \frac{z}{-1} = \frac{\cos\theta + i\sin\theta}{(1 - \cos\theta) - i\sin\theta} \times \frac{(1 - \cos\theta) + i\sin\theta}{(1 - \cos\theta) + i\sin\theta}$$

$$= \frac{\cos\theta - \cos^2\theta + i\sin\theta\cos\theta + i\sin\theta - i\sin\theta\cos\theta - \sin^2\theta}{(1 - \cos\theta)^2 + \sin^2\theta}$$

$$= \frac{\cos\theta - (\cos^2\theta + \sin^2\theta) + i\sin\theta}{(1 - \cos\theta)^2 + \sin^2\theta}$$

$$\frac{z}{-1} = \frac{(\cos\theta - 1) + i\sin\theta}{1 - 2\cos\theta + \cos^2\theta + \sin^2\theta} = \frac{(\cos\theta - 1) + i\sin\theta}{2 - 2\cos\theta}$$

$$z = \frac{(\cos\theta - 1) + i\sin\theta}{2\cos\theta - 2} = \frac{\cos\theta - 1 + i\sin\theta}{2(\cos\theta - 1)}$$

$$= \frac{1}{2} + \frac{i\sin\theta}{(\cos\theta - 1)}$$

$$= \frac{1}{2} + \frac{i \cancel{2\sin\theta/2} \cos\theta/2}{(-2\sin^2\theta/2)} = \frac{1}{2} - i \cot\left(\frac{\theta}{2}\right)$$